MA334 Assignment Report

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1.Data Exploration

(a). Summary of the variables in data set

```
## 'data.frame':    1000 obs. of  10 variables:
##  $ carat  : num  0.82 1.02 0.32 0.53 0.3 0.41 0.53 0.36 0.85 0.4 ...
##  $ cut    : chr  "Ideal" "Ideal" "Very Good" "Very Good" ...
##  $ color  : chr  "F" "E" "F" "E" ...
##  $ clarity: chr  "VS2" "IF" "VS1" "VVS2" ...
##  $ depth  : num  61.9 62.6 62.3 60.7 62.6 61.5 60.5 62.3 61.6 61.5 ...
##  $ table  : num  57 57 55 59 54 55 57 56 54 56 ...
##  $ price  : int  3514 11500 696 2297 518 763 2542 1053 4164 1158 ...
##  $ x      : num  5.96 6.41 4.38 5.19 4.24 4.78 5.23 4.56 6.08 4.74 ...
##  $ y      : num  5.99 6.47 4.42 5.28 4.29 4.81 5.31 4.53 6.12 4.79 ...
##  $ z      : num  3.7 4.03 2.74 3.18 2.67 2.95 3.19 2.83 3.76 2.93 ...
```

The dataset contains 1,000 observations and 10 variables. The variables are categorized as follows:

Table 1: Summary of Variables in the Dataset

Variable.Type	Count	Names
Quantitative	7	Carat, Depth, Table, Price, x, y, z
Qualitative	3	Cut, Color, Clarity

Identify categories for qualitative variables:

Cut:

```
##
##      Fair      Good     Ideal   Premium Very Good
##        26        95       400       263       216
```

Color:

```
##
##   D   E   F   G   H   I   J
## 121 180 178 198 174 104  45
```

Clarity:

```
##
##   I1   IF  SI1  SI2  VS1  VS2 VVS1 VVS2
##   12   31  212  190  155  233   75   92
```

Based on the frequency tables above, the categories for the qualitative variables are as follows:

- Cut: Ideal, Very Good, Premium, Good, and Fair
- Color: D, E, F, G, H, I, and J
- Clarity: IF, VVS1, VVS2, VS1, VS2, SI1, SI2, and I1

(b). Location and Spread of Variables

Quantitative Variables:

Table 2: Descriptive Statistics for Quantitative Variables

	Mean	Median	Mode	SD	Min	Max	Range
carat	0.80	0.70	0.30	0.49	0.23	4.01	3.78
depth	61.75	61.80	61.60	1.44	54.20	69.70	15.50
table	57.45	57.00	56.00	2.19	49.00	67.00	18.00
price	3935.48	2376.00	776.00	4036.14	371.00	18729.00	18358.00
x	5.74	5.68	4.37	1.13	3.90	10.02	6.12
y	5.74	5.69	4.34	1.12	3.92	9.94	6.02
z	3.54	3.52	2.72	0.70	2.41	6.24	3.83

Qualitative Variables:

Table 3: Location and Spread for Qualitative Variables

Variable	Location (Mode)	Spread (Categories)
Cut	Ideal	5 Categories
Color	G	7 Categories
Clarity	VS2	8 Categories

Explanation of Location and Spread:

- Price Skewness: The Mean is significantly higher than the Median, indicating a positive (right) skew.This suggests that while most diamonds are priced lower, a few high-value outliers are pulling the average upward.
- Variable Spread: The high Standard Deviation for price and a High Range highlight extreme price variability.
- Most common diamond characteristic: For qualitative data, the mode identifies the most frequent characteristics are Ideal cut, G color, and VS2 clarity.

(c). Visualizations for the distributions of variables

Histograms for quantitative variables:

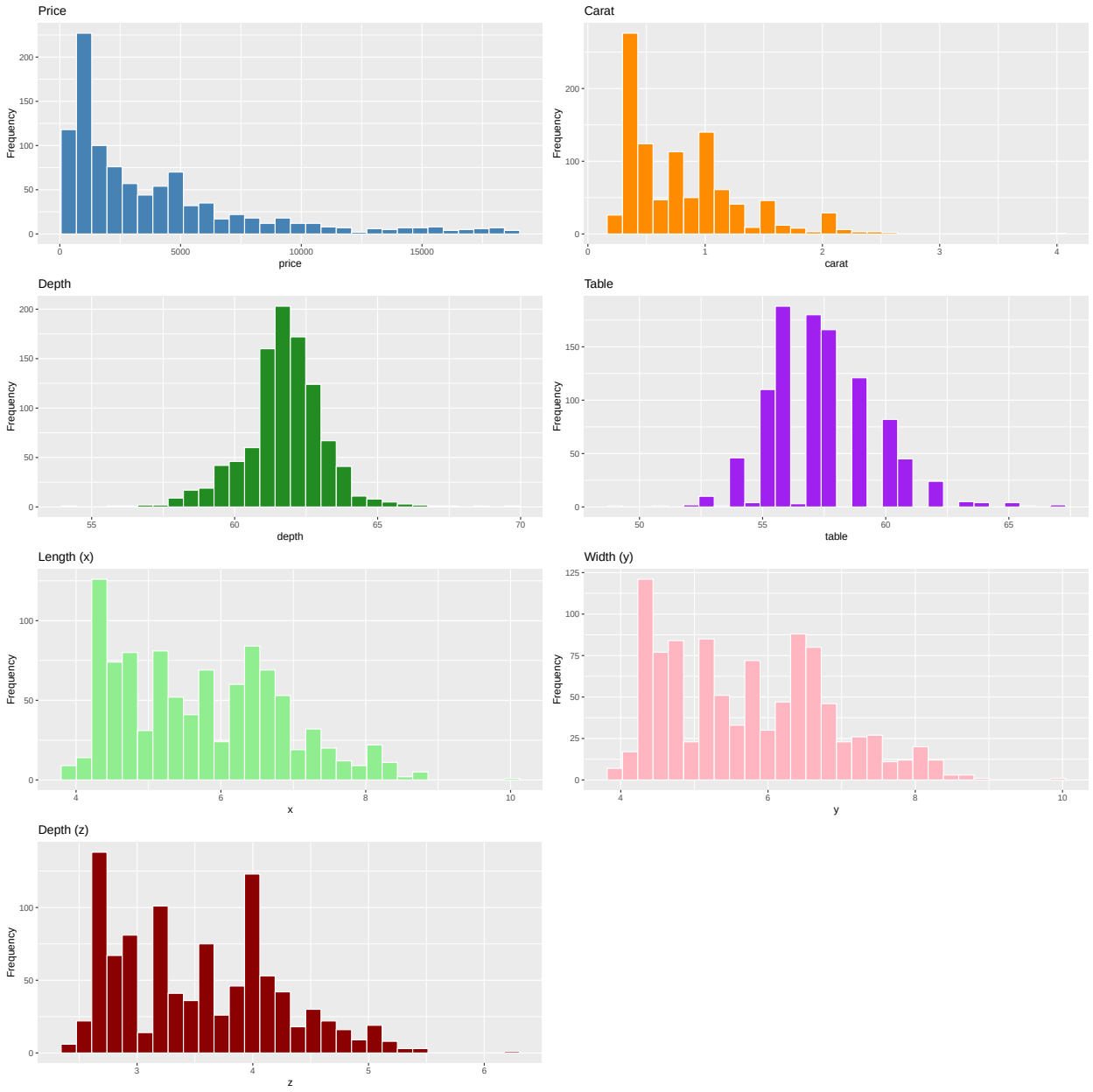


Figure 1: Distribution of Quantitative Variables

Bar charts for qualitative variables:

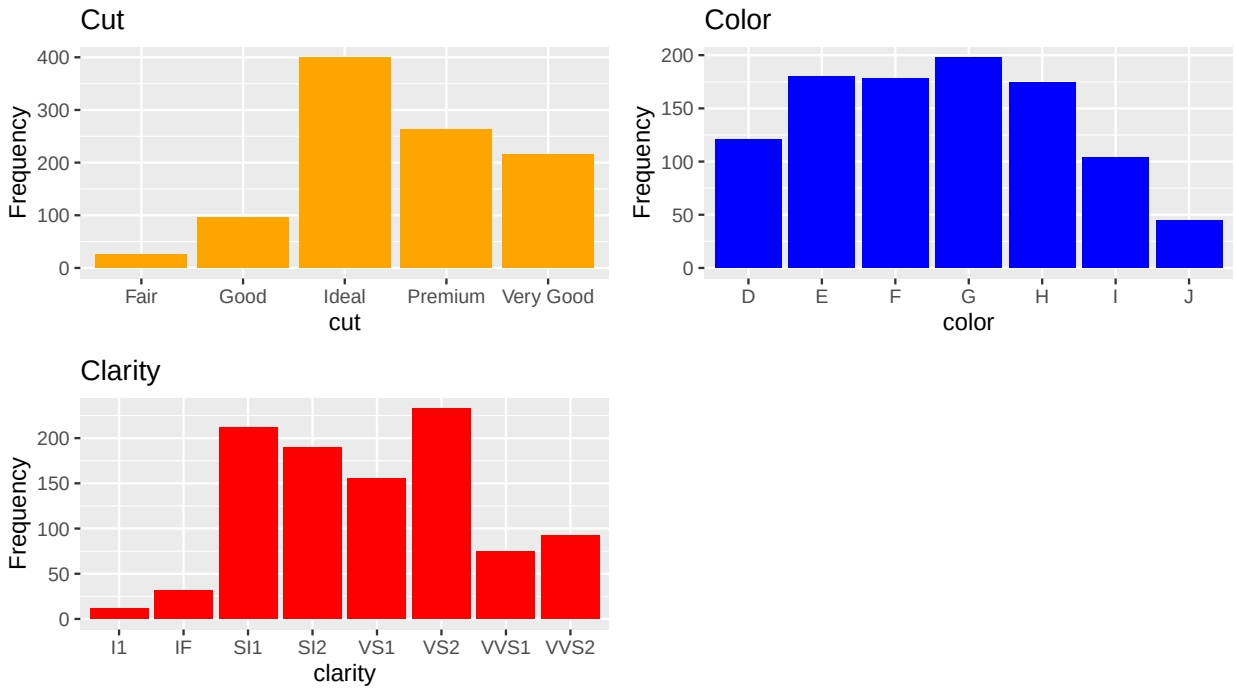


Figure 2: Distribution of Qualitative Variables

(d). Calculate correlation matrix between the numeric variables and find out which variables are strongly correlated with price

Table 4: Correlation Matrix of Numeric Variables

	carat	depth	table	price	x	y	z
carat	1.00	0.05	0.21	0.92	0.98	0.97	0.97
depth	0.05	1.00	-0.27	0.00	-0.01	-0.01	0.11
table	0.21	-0.27	1.00	0.15	0.21	0.21	0.18
price	0.92	0.00	0.15	1.00	0.89	0.89	0.89
x	0.98	-0.01	0.21	0.89	1.00	1.00	0.99
y	0.97	-0.01	0.21	0.89	1.00	1.00	0.99
z	0.97	0.11	0.18	0.89	0.99	0.99	1.00

Based on the observed dataset, the correlation matrix shows a very strong relationship between Carat and Price ($r = 0.92$). This confirms that as the weight of the diamond increases, the price tends to rise significantly. This is followed closely by the physical dimensions (x, y, and z), which show identical high correlations with price (r = 0.89). In contrast, variables such as Table (r = 0.15) and Depth (r = 0.00) show almost no correlation with price.

2. Probability, probability distributions and confidence intervals

(a). Probability Calculations

i. Probability price exceeds $10,000.

```
## [1] 0.09
```

The analysis shows that the probability of a randomly chosen diamond's price exceeding $10,000 is 9%.

ii. Probability that a diamond is of Ideal cut.

```
## [1] 0.4
```

The analysis shows that the probability of a randomly chosen diamond being an Ideal cut is 40%.

(b). From a random sample of 20 diamonds, calculate the probability that more than 12 of them are Ideal cut.

```
## [1] 0.02
```

The analysis shows that in a random selection of 20 diamonds, there is a 2% probability that more than 12 will possess an 'Ideal' cut.

(c). Construct both 90% and 95% confidence intervals for the mean carat weight. See which interval is wider.

#The formula for a Confidence Interval (CI) is $\bar{x} \pm z \cdot \frac{s}{\sqrt{n}}$, where $\bar{x}$ is the sample mean, s is the standard deviation, and n is the sample size.

90% CI:

```
## [1] 0.7757567 0.8264233
```

95% CI:

```
## [1] 0.7709036 0.8312764
```

The 95% confidence interval is wider because increasing the confidence level requires a larger critical value ($z \approx 1.96$ for 95% vs. $z \approx 1.645$ for 90%). To be more certain that the interval contains the true population mean, the "margin of error" must be increased, resulting in a broader range.

3. Hypothesis Tests

(a). Visualization

i&ii.Visualizing the Price Distribution: Ideal and Very Good cuts

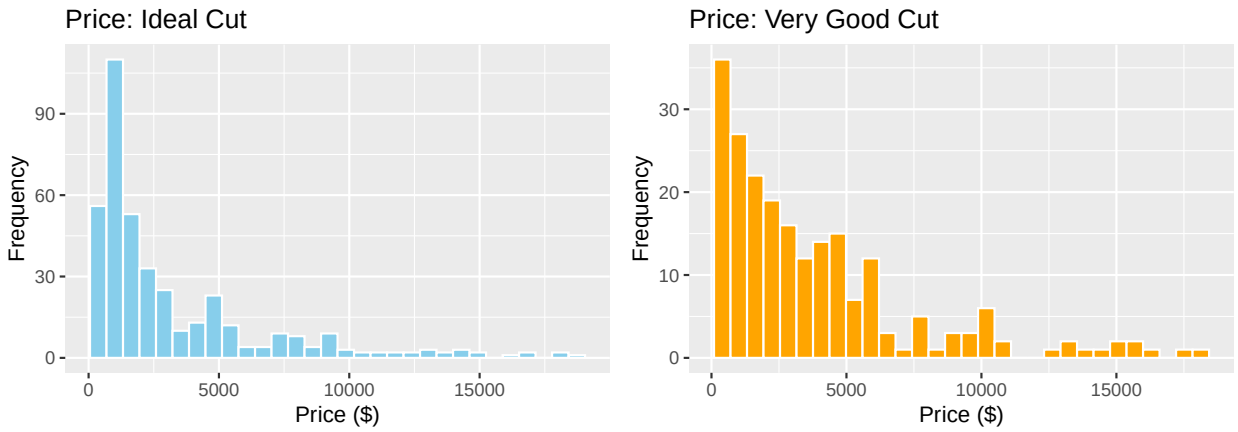


Figure 3: Comparison of Price Distributions of Ideal and Very Good Cut

iii.Comparison of Histogram Distributions

Based on the histograms, the data appears to be right-skewed (positively skewed) with a long tail extending toward the higher prices. Therefore, it is not normally distributed.

iv.Create a new variable called ln(price) which is the logarithm transformation of price. Repeat the steps i-iii, and determine whether the ln(price) values for diamonds with an 'Ideal' or 'Very Good' cut appear more normally distributed.

Log Transformation

Visualizing the ln(price) Distribution: Ideal and Very Good Cuts

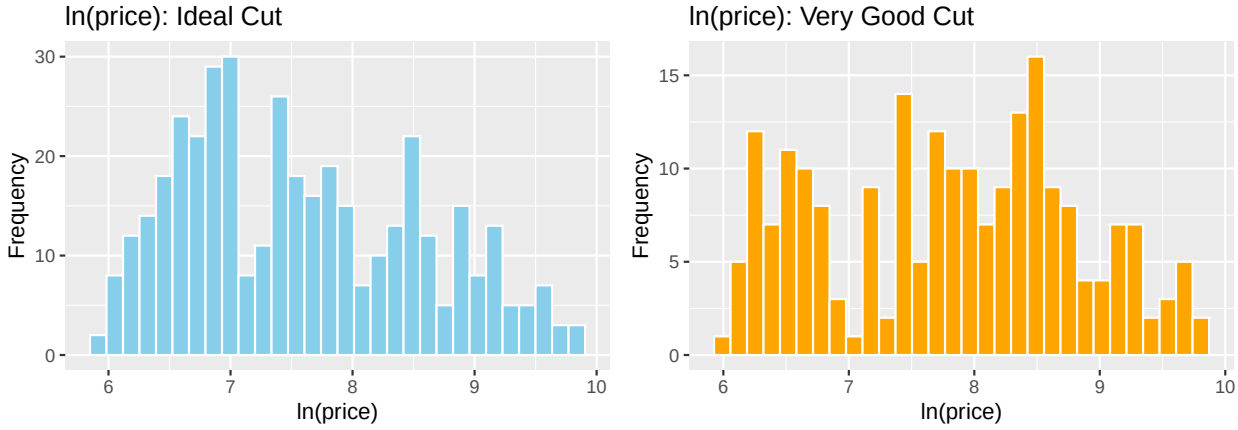


Figure 4: Comparison of ln(price) Distributions of Ideal and Very Good Cut

Base on the histograms after applying the logarithm transformation, the distributions for both 'Ideal' and 'Very Good' cuts appear much more normally distributed. The histograms have lost their long right-side tails and now show a more symmetric.

(b). T-test on Raw Price (Ideal vs. Very Good)

State the null and alternative hypotheses:

$H_0 : \mu_{Ideal} = \mu_{VeryGood}$ (The average price is the same for both groups)

$H_1 : \mu_{Ideal} \neq \mu_{VeryGood}$ (The average price is different between the two groups)

Decide the significance level: $\alpha = 0.05$

Formulate a test criterion: Reject H_0 if p-value $< \alpha$

Calculate using two sample t-test:

```
##
##  Welch Two Sample t-test
##
## data:  price by cut
## t = -2.0765, df = 421.11, p-value = 0.03845
## alternative hypothesis: true difference in means between group Ideal and group Very Good is not equal to 0
## 95 percent confidence interval:
##  -1268.22604   -34.79951
## sample estimates:
##     mean in group Ideal mean in group Very Good
##                3227.015                3878.528
```

The calculated p-value is 0.03845, which is less than the significance level $\alpha = 0.05$ Therefore, we reject the null hypothesis (H_0) in favor of the alternative (H_1). There is sufficient evidence to suggest that the mean price of 'Ideal' diamonds is significantly different from 'Very Good' diamonds.

(c). T-test on ln(price) (Ideal vs. Very Good)

State the null and alternative hypotheses:

$H_0 : \mu_{Ideal} = \mu_{VeryGood}$ (The average price is the same for both groups)

$H_1 : \mu_{Ideal} \neq \mu_{VeryGood}$ (The average price is different between the two groups)

Decide the significance level: $\alpha = 0.05$

Formulate a test criterion: Reject H_0 if p-value $< \alpha$

Calculate using two sample t-test:

```
##
##  Welch Two Sample t-test
##
## data:  ln(price) by cut
## t = -2.6224, df = 430.65, p-value = 0.009041
## alternative hypothesis: true difference in means between group Ideal and group Very Good is not equal to 0
## 95 percent confidence interval:
##  -0.38327274 -0.05487786
## sample estimates:
##     mean in group Ideal mean in group Very Good
##                7.586022                7.805098
```

The calculated p-value is 0.009041, which is less than the significance level $\alpha = 0.05$. Therefore, we reject the null hypothesis (H_0) in favor of the alternative (H_1). There is sufficient evidence to suggest that the mean $ln(price)$ of 'Ideal' diamonds is significantly different from 'Very Good' diamonds.

(d). Comparing Conclusions from (b) and (c)

Both tests lead to the rejection of the null hypothesis ($p < 0.05$), confirming a significant price difference between "Ideal" and "Very Good" cuts. However, the results from 3(c) are more robust. While the raw price data in 3(a) was

heavily right skewed violating the normality assumption the log-transformation successfully corrected this skewness, making the second t-test more statistically valid.

(e). Independence test: Relationship Between Price Level and Clarity

Convert the price into a binary variable: high price ($> \$5{,}000$) vs low price ($\leq \$5{,}000$)

State the null and alternative hypotheses:

H_0: Price level and clarity level are independent. (This means clarity does not affect whether a diamond is expensive).

H_1: Price level and clarity level are dependent. (This means there is a relationship between clarity and price category)

Decide the significance level: $\alpha = 0.05$

Formulate a test criterion: Reject H_0 if p-value $< \alpha$

Create Contingency Table:

Table 5: Contingency Table of Price vs Clarity

	I1	IF	SI1	SI2	VS1	VS2	VVS1	VVS2
High	3	3	65	53	37	68	14	18
Low	9	28	147	137	118	165	61	74

Calculate using Chi-Square test:

```
##
##  Pearson's Chi-squared test
##
## data:  price_clarity_table
## X-squared = 12.679, df = 7, p-value = 0.08031
```

Based on the Chi-Square test result, the p-value is 0.08031, which is greater than the significance level $\alpha = 0.05$. Therefore, we fail to reject the null hypothesis (H_0). There is insufficient evidence to suggest that price category and clarity level are dependent. This means we conclude that clarity level does not have a statistically significant effect on whether a diamond's price exceeds $5,000.

4. Linear Regression

(a). Scatter plot of ln(price) vs. ln(carat)

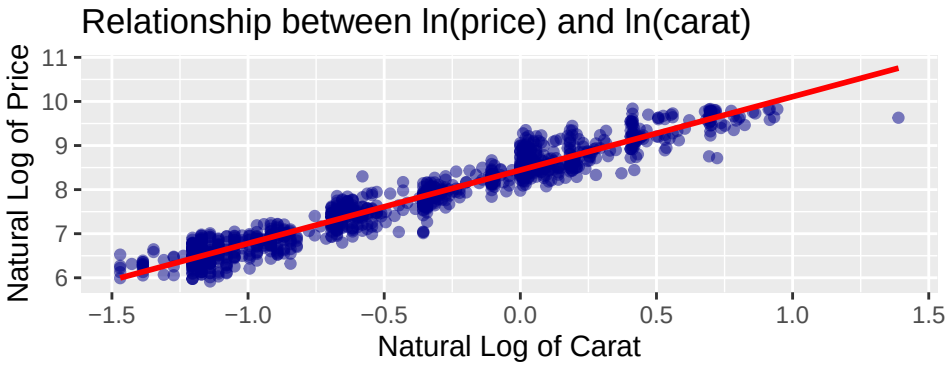


Figure 5: Scatter Plot of Log-Price vs. Log-Carat

(b). Fit a simple linear regression model with ln(price) as the response and ln(carat) as the predictor

Linear Regression model:

```
##
## Call:
## lm(formula = `ln(price)` ~ `ln(carat)`, data = diamond_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.12566 -0.16576 -0.00552  0.16084  0.87329
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 8.443842   0.009848   857.4   <2e-16 ***
## `ln(carat)` 1.665033   0.013942   119.4   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2586 on 998 degrees of freedom
## Multiple R-squared:  0.9346, Adjusted R-squared:  0.9345
## F-statistic: 1.426e+04 on 1 and 998 DF,  p-value: < 2.2e-16
```

i.Report the estimated slope and interpretation

Base on Linear Regression Model the estimated slope is 1.665 therefore its indicates that for a 1% increase in the carat (weight) of a diamond, the price is estimated to increase by approximately 1

ii.Test whether carat is a significant predictor

To test if carat is a significant predictor, we use the following hypotheses:

Null Hypothesis (H_0): $\beta_1 = 0$ (Carat weight has no effect on price)

Alternative Hypothesis (H_1): $\beta_1 \neq 0$ (Carat weight is a significant predictor of price)

Base on summary of the linear regression model ln(carat) p-values is less than 2e-16. Since this p-value is significantly less than $\alpha = 0.05$, we reject the null hypothesis (H_0). This provides strong statistical evidence that carat weight is a significant predictor of diamond price.

iii.Report the coefficient of determination

Base on summary of the linear regression model the coefficient of determination (R^2) is 0.9346. This means that approximately 93.46% of the total variation in the response variable $ln(price)$ is explained by the linear relationship with the predictor variable $ln(carat)$.

(c). Multiple Linear Regression: ln(price) and ln(carat) + cut

Multiple Linear Regression model:

```
##
## Call:
## lm(formula = `ln(price)` ~ `ln(carat)` + cut, data = diamond_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.14479 -0.16096 -0.00611  0.15924  0.91853
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   8.22078    0.04971 165.360  < 2e-16 ***
## `ln(carat)`   1.67991    0.01399 120.089  < 2e-16 ***
## cutGood       0.12160    0.05600   2.172   0.0301 *
```

```
## cutIdeal      0.27124    0.05139   5.278 1.60e-07 ***
## cutPremium    0.22153    0.05201   4.260 2.24e-05 ***
## cutVery Good  0.23427    0.05256   4.457 9.26e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.253 on 994 degrees of freedom
## Multiple R-squared:  0.9377, Adjusted R-squared:  0.9373
## F-statistic:  2990 on 5 and 994 DF,  p-value: < 2.2e-16
```

Estimated regression model formula is $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots$

Base on our multiple linear regression model the estimated model is:

$\widehat{ln(Price)} = 8.22078 + 1.67991 \cdot ln(Carat) + 0.12160 \cdot Cut_{Good} + 0.27124 \cdot Cut_{Ideal} + 0.22153 \cdot Cut_{Premium} + 0.23427 \cdot Cut_{VeryGood}$

Note: The 'Fair' cut is included in the Intercept. The coefficients for other cuts represent the price difference relative to a 'Fair' cut diamond.

(d). State which variables in Question (c) are statistically significant at the 1% level

Base on multiple linear regression model we obtained p-values of all variables as below:

- ln(carat): $< 2e-16$
- cutGood: 0.0301
- cutIdeal: $1.60e-07$
- cutPremium: $2.24e-05$
- cutVery Good: $9.26e-06$

To be significant at the 1% level, the p-value must be less than 0.01.

Significant Variables: $ln(carat)$, Cut_{Ideal}, $Cut_{Premium}$, and $Cut_{VeryGood}$ all have p-values < 0.01 and are therefore statistically significant at the 1% level.

Non-Significant Variable: The variable Cut_{Good} has a p-value of 0.0301. Since $0.0301 > 0.01$, Cut_{Good} is not statistically significant at the 1% level (though it would be significant at the 5% level).

(e). Hypothesis Test for the Significance of ln(carat) in the Multiple Regression Model

To determine if ln(carat) remains a significant predictor while accounting for the effect of cut, we perform an individual t-test on its regression coefficient.

State the null and alternative hypotheses:

Null Hypothesis (H_0): $\beta_{ln(carat)} = 0$. This assumes that $ln(carat)$ has no linear relationship with $ln(price)$ when the diamond's cut is included in the model

Alternative Hypothesis (H_1): $\beta_{ln(carat)} \neq 0$. This assumes that $ln(carat)$ is a significant predictor of $ln(price)$ even after controlling for the diamond's cut

(f). Conduct the hypothesis testing in (e) and report the result

To conduct this test, we examine the t-statistic and the corresponding p-value for $ln(carat)$ from the multiple regression output generated in Question 4(c).

Test Results:

- **p-value:** From the R summary, the p-value for ln(carat) is $<$ 2e-16.
- **Significance Level:** $\alpha = 0.01$
- **Conclusion:** Since the p-value $<$ 2e-16 is much smaller than the significance level of 0.01, we reject the null hypothesis (H_0). There is overwhelming statistical evidence to conclude that the weight of a diamond ($ln(carat)$) is a highly significant predictor of its price, even when the quality of the cut is taken into account.